Haoyang Zheng

PH.D. CANDIDATE AT PURDUE UNIVERSITY | MACHINE LEARNING & AI RESEARCHER

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Education

Purdue University, College of Engineering — *West Lafayette, IN*
Ph.D. in Mechanical Engineering (advised by Prof. Guang Lin, GPA 4.0/4.0) — *Jun. 2021 - Dec. 2026 (Expected)*

Purdue University, College of Engineering — *West Lafayette, IN*
M.S. in Mechanical Engineering (GPA 3.8/4.0) — *Sep. 2019 - May. 2021*

Southwest University, School of Computer and Information Science — *Chongqing, China*
B.Eng. in Automation (advised by Prof. Yong Deng, Rank 1/92) — *Sep. 2014 - Jun. 2018*

Professional Experience

Google Research — *Mountain View, CA*
Google Research Intern — *May. 2026 – Aug. 2026 (Expected)*

- Developed novel algorithms to accelerate inference for **Diffusion Language Models** in on-device generative AI settings.
- Analyzed trade-offs among inference latency, memory footprint, and model accuracy under edge-device constraints.
- Benchmarked proposed methods on real-world, latency-sensitive generation tasks.

Xiaohongshu (RedNote) Inc, Humane Intelligence Lab (hi lab) — *Beijing, China*
ACE Intern Program — *Dec. 2025 – Apr. 2026*

- Contributed to the **SFT data distillation pipeline** for a **multimodal image-editing model** by building multi-GPU **OCR** and recaptioning workflows over datasets, extracting text boxes and transcripts with **HunyuanOCR** and enriching samples with **Qwen3-VL** captions.
- Built teacher-assisted data generation workflows that synthesized bilingual add/modify/remove editing instructions from OCR outputs and used **Qwen-Image-Edit** and **Nano Banana** to generate edited images at scale with task-specific prompts and distributed processing.
- Designed automated quality control for distilled samples using taxonomy-based corner/bad-case tagging, OCR- derived edit masks, **LPIPS**, **DINOv2**, **MUSIQ**, and **Qwen**-based instruction-execution checks, then validated the resulting data through **Streamlit** QA tools and post-training evaluations on pretrained checkpoints.

Argonne National Laboratory, Mathematics and Computer Science Division — *Lemont, IL*
Givens Associate — *May. 2023 - Jul. 2023*

- Proposed scalable **Proximal Policy Optimization (PPO)** algorithms integrated with Bayesian optimization in control tasks.
- Developed AI-enhanced modeling and simulation pipelines for physical systems by combining machine learning with PDE solvers.

Research Interests

I study **Generative Modeling** and **Reinforcement Learning** (RL) from a probabilistic modeling perspective. My interests include **LLMs**, continuous and discrete diffusion, RL, and sampling/optimization methods that improve latency and robustness.

My recent work on **discrete diffusion distillation** achieves over **60**$\times$ faster language generation than GPT baseline with similar quality. This research direction naturally extends to a range of emerging domains, including multimodal diffusion, code generation, reasoning models, etc. I am also interested in scaling these methodologies to larger LLM and MoE architectures, where improving efficiency and stability remains a central challenge. My long-term goal is to enable next-generation large-scale models that achieve both extreme efficiency and the reliability needed for practical, real-world use.

Papers

Large Language Models

[1] **Haoyang Zheng**, Xinyang Liu, Xiangrui Kong, Nan Jiang, Zheyuan Hu, Weijian Luo*, Wei Deng*, Guang Lin, "Ultra-Fast Language Generation via Discrete Diffusion Divergence Instruct", *ICLR 2026*.

Reinforcement Learning

[1] **Haoyang Zheng**, Wei Deng*, Christian Moya, Guang Lin*, "Accelerating Approximate Thompson Sampling with Underdamped Langevin Monte Carlo", *AISTATS 2024*.

[2] Weixin Wang, **Haoyang Zheng**, Guang Lin, Wei Deng, Pan Xu*, "Rethinking Langevin Thompson Sampling from A Stochastic Approximation Perspective", *NeurIPS 2025 Workshop on Dynamics at the Frontiers of Optimization, Sampling, and Games (DynaFront)*.

Sampling, Markov-Chain Monte Carlo

[1] **Haoyang Zheng**, Ruqi Zhang*, Guang Lin*, "Exploring Non-Convex Discrete Energy Landscapes: A Langevin-Like Sampler with Replica Exchange", *AAAI 2026 (Oral Presentation)*.

[2] **Haoyang Zheng**, Hengrong Du, Qi Feng, Wei Deng*, Guang Lin*, "Constrained Exploration via Reflected Replica Exchange Stochastic Gradient Langevin Dynamics", *ICML 2024*.

[3] Cindy Xiangrui Kong, **Haoyang Zheng**, Guang Lin, "LAPD: Langevin-Assisted Bayesian Active Learning for Physical Discovery", *arXiv:2503.02983* (2025).

AI for Science

[1] **Haoyang Zheng**, Jeffrey R. Petrella, P. Murali Doraiswamy, Guang Lin*, Wenrui Hao, "Data-Driven Causal Model Discovery and Personalized Prediction in Alzheimer's Disease", *Nature Digital Medicine* (2022).

[2] Yikai Liu, **Haoyang Zheng**, Lining Mao, Yanbin Wang, Ming Chen, Guang Lin, "ProTDyn: a foundation Protein language model for Thermodynamics and Dynamics generation", *ICLR 2026*.

[3] Cindy Xiangrui Kong, Yueqi Wang, **Haoyang Zheng**, Weijian Luo, Guang Lin, "Ultra Fast PDE Solving via Physics Guided Few-step Diffusion", *arXiv:2602.03627* (2026).

[4] **Haoyang Zheng**, Guang Lin*, "LES-SINDy: Laplace-Enhanced Sparse Identification of Nonlinear Dynamical Systems", *Journal of Computational Physics* (2025).

[5] **Haoyang Zheng**, Yao Huang, Ziyang Huang, Wenrui Hao, Guang Lin*, "HomPINNs: Homotopy Physics-Informed Neural Networks for Inverse Problems", *Journal of Computational Physics* (2024).

[6] **Haoyang Zheng**, Ziyang Huang, Guang Lin, "A physics-constrained neural network for multiphase flows", *Physics of Fluids* (2022).

[7] **Haoyang Zheng**, Guang Lin, "Muti-Fidelity Prediction and Uncertainty Quantification with Laplace Neural Operators for Parametric Partial Differential Equations", *arXiv:2502.00550* (2025).

Decision Systems

[1] **Haoyang Zheng**, Yong Deng, Yong Hu, "Fuzzy evidential influence diagram and its evaluation algorithm", *Knowledge-Based Systems* (2017).

[2] **Haoyang Zheng**, Yong Deng, "Evaluation method based on fuzzy relations between Dempster–Shafer belief structure", *International Journal of Intelligent Systems* (2018).

[3] Likang Yin, **Haoyang Zheng**, Tian Bian, Yong Deng, "An evidential link prediction method and link predictability based on Shannon entropy", *Physica A: Statistical Mechanics and its Applications* (2017).

[4] Tian Bian, **Haoyang Zheng**, Likang Yin, Yong Deng, "Failure mode and effects analysis based on D numbers and TOPSIS", *Quality and Reliability Engineering International* (2018).

Talks

[1] On Sampling Tasks with Langevin Dynamics, Brown University, July 2024.
Host: George Em Karniadakis.

[2] Ultra-Fast Language Generation via Discrete Diffusion Divergence Instruct, Peking University, January 2026. Host: He Sun.

[3] Ultra-Fast Language Generation via Discrete Diffusion Divergence Instruct, Brown University, January 2026. Host: George Em Karniadakis.

[4] Exploring Non-Convex Discrete Energy Landscapes: A Langevin-Like Sampler with Replica Exchange, January 2026, on AAAI 2026, at Singapore

Posters

[1] Reflected Replica Exchange Stochastic Gradient Langevin Dynamics, September 2024, on Mathematical and Scientific Foundations of Deep Learning Annual Meeting (Simons Foundation), at New York, NY.

[2] Accelerating Thompson Sampling with Underdamped Langevin Monte Carlo, October 2024, on SIAM Conference on Mathematics of Data Science, at Atlanta, GA.

[3] On Thompson Sampling with Underdamped Langevin Monte Carlo, September 2023, on Mathematical and Scientific Foundations of Deep Learning Annual Meeting (Simons Foundation), at New York, NY.

Achievements

Selected Awards

2025	**Student Travel Awards**, Simons Foundation, 2025 MoDL	*New York, NY*
2024	**Student Travel Awards**, Simons Foundation, 2024 MoDL	*New York, NY*
2024	**Student Travel Awards**, SIAM Conference on MDS24	*Atlanta, GA*
2024	**Student Travel Awards**, Nonlocality: Challenges in Modeling and Simulation	*Providence, RI*
2023	**Student Travel Awards**, Simons Foundation, 2023 MoDL	*New York, NY*
2018	**Pacemaker to Technological Innovation in Chongqing**, awarded 10 undergraduate every two years	*Chongqing, China*
2018	**Outstanding undergraduates in Chongqing**, awarded 1% undergraduate every year	*Chongqing, China*
2015	**China National Scholarship**, awarded 1% undergraduates every year	*China*

Selected Honors

2017	**Finalist (0.5%)**, Interdisciplinary Contest in Modeling	*China*
2016	**Special prize (2/3568)**, International Mathematical Contest in Modeling	*China*

Service

Conference Reviewer	ICLR, ICML, NeurIPS, AISTATS, KDD, AAAI
Journal Reviewer	IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Cybernetics (TCYB), IEEE Transactions on Multimedia (TMM), Transactions on Machine Learning Research (TMLR), SIAM Multiscale Modeling and Simulation, Journal of Computing and Information Science in Engineering, Journal of Supercomputing, Scientific Reports, Knowledge and Information Systems

Skills & Tools

Machine Learning	Generative Modeling, Reinforcement Learning, Bayesian Optimization, Sampling
ML/DL Frameworks	PyTorch, DDP, FSDP2, vllm, OpenAI Gym, HuggingFace
Programming Languages	Python, Bash, MATLAB
Uncertainty Quantification	Bayesian Inference, Langevin Dynamics, MCMC